MSc Robotics

ME5415 Advanced Soft Robotics
Module briefing

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Module syllabus and schedule

Week	Date (Thu.)	Topic	Lecturer	Assessment
1	15/01/26	Module briefing + introduction to origami robotics	Hongying Zhang	
2	22/01/26	Introduction to soft robotics **Paper survey assignments + project assignments**	Cecilia Laschi	
3	29/01/25	Soft Actuation Technologies	Cecilia Laschi	
4	05/02/26	Modelling and control of soft robots	Cecilia Laschi	
5	12/02/26	Typical origami patterns	Hongying Zhang	
6	19/02/26	Assignment time		No class
Recess				
7	05/03/26	Mid-term quiz	Students	CA 1 (30%)
8	12/03/26	Modeling of origami robots	Hongying Zhang	
9	19/03/26	Tutorial on pneumatic actuation	Hongying Zhang	
10	26/03/26	Survey report submission	Students	CA 2 (20%)
11		Project time and lecturers' consultation (upon booking)		
12		Project time and lecturers' consultation (upon booking)		
13	16/04/26	Project report submission (paper-like)	Students	CA 3 (50%)
Reading				
Exam.				

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Journal paper assignments (assessed, 20%)

- Groups of 4 students
- Please choose one of the topics from the list given (next slide)*, per each group**
- Make a literature survey on the topic
  - Journal papers from top international journals are preferable.
- Write a report in the form a review paper on the topic and submit it in Canvas
  - The level of critical discussion of the papers surveyed contributes to the final grade. Keep the report within 6 pages.
- Report submission on March 27th

*You may propose a different topic, but please check it with the lecturers
**The same topic can be chosen by more than 1 group.

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Journal paper assignments (assessed, 20%)

List of topics:

- EAP-based soft robots
- SMA-actuated soft robots
- DE-actuated soft robots
- HASEL actuators
- Inflatable robots
- Soft sensors for tactile sensing
- Soft stretch/strain sensors
- Textile-based sensors
- EIT sensors
- Chemically-actuated soft robots
- Growing soft robots
- Self-healing soft robots
- CC/PCC models for soft robots
- Learning-base soft robot control
- Model-based soft robot control
- Reinforcement learning in soft robotics
- Soft robots in endoscopy
- Underwater soft robots
- Wearable soft robots
- Reconfigurable/Shape-Changing Origami Robots
- Magnetic-driven Untethered Origami Robots
- Self-folding Origamis
- Deployable Origami in Aerospace
- (Electronic-free) Fluidic Controllers for Soft Robot

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Project assignments (assessed, 50%)

- Groups of 4 students
- Two tasks to choose from:
  - food handling for tray preparation
  - in-pipe locomotion
- or EMIO soft robotic kit
- Each group free to choose different strategy, design, technologies
- Prototypes not expected, but design/simulation
- Report in the format of a 6-page research paper
  - Submission: April 30th

Read more

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